

1. Scale down (to reduce the voltage or frequency)
2. Cut off (completely disconnect the voltage or frequency)

There are different methods with which you can take care of power management factors in CE devices. They are:

- Dynamic voltage and frequency scaling (DVFS)
- Tickless kernel
- Power Top (power consumption monitoring tool)
- Suspend/resume
- Device power management
- Dynamic power management

Dynamic voltage and frequency scaling

The bottom line is that power dissipation can be reduced if we reduce or cut off the voltage or frequency. There are certain preset values for the voltage, current and frequency for each device, device state and set of inputs. This is called the operating point (OPP). This operating point can be dynamically set when the device is working. This is called DVFS (dynamic voltage and frequency scaling). The option to enable frequency scaling is shown in Figure 3. In fact, DVFS is about dynamically changing the operating point (OPP) to suit execution environments [3].

The tickless kernel

The system produces a timer interrupt every 1 ms (if the timer frequency is set to 1000 Hz), and when this interrupt occurs, the CPU stops what it is doing and executes the timer interrupt handler function. In the interrupt handler function, there are different tasks to be done, such as checking if there is any other higher priority task to be done [10].

On an idle system, the processor wakes up every time a timer interrupt is received, only to find that there is nothing else to do. Can we disable the timer interrupt when all the processors are idle, and enable it only when there is something to be done or when there is an interrupt?

If we do this, then it is called a tickless kernel. If there is less work to do in the device, the tickless kernel sends the processor into sleep mode whenever it's possible, and this reduces power consumption. Figure 4 shows the option to enable the tickless kernel feature.

There is reported data available through the ELF Conference about a machine working on Intel Core Duo at 1000 Hz. A comparison was made of the CPU idle residency — with and without ticks. It was found that when the number of interrupts was reduced by 20 times, the CPU idle time increased by 10 times, which would result in considerable power savings.

Power Top

Power Top [11] is a utility that can give you the top 10 power consuming tasks in the system. It is only a power consumption monitoring tool to guide you to optimise the application that is consuming more power and bring the overall power consumption under control. Figures 5 and 6 show the output of the Power Top tool when a laptop is running on battery power supply and when it is connected to the main power supply, respectively.

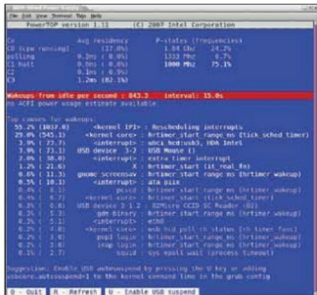


Figure 6: Power Top output when a laptop runs on the main power supply



Figure 7: Option to enable suspend to disk and resume

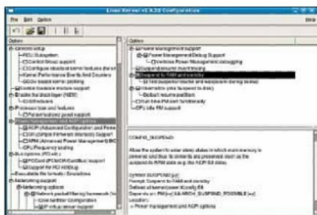


Figure 8: Option to enable suspend to RAM and resume

Suspend and resume

The kernel provides hooks and device drivers to support suspend/resume operations when there are long periods of idle